Hy-Tracker: A Novel Framework for Enhancing Efficiency and Accuracy of Object Tracking in Hyperspectral Videos

Mohammad Aminul Islam, Wangzhi Xing, Jun Zhou, *Senior Member, IEEE*, Yongsheng Gao, *Senior Member, IEEE*, and Kuldip K. Paliwal

***Abstract*— Hyperspectral images, with their many spectral bands, provide a rich source of material information about an object that can be effectively used for object tracking. However, many trackers in this domain rely on detection-based techniques, which often perform suboptimally in challenging scenarios such as managing occlusions and distinguishing objects in cluttered backgrounds. This underperformance is primarily due to the presence of multiple spectral bands and the inability to leverage this abundance of data for effective tracking. Additionally, the scarcity of annotated hyperspectral videos and the absence of comprehensive temporal information exacerbate these difficulties, further limiting the effectiveness of current tracking methods. To address these challenges, this article introduces the novel Hy-Tracker framework, designed to bridge the gap between hyperspectral data and state-of-the-art object detection methods. Our approach leverages the strengths of YOLOv7 for object tracking in hyperspectral videos, enhancing both accuracy and robustness in complex scenarios. The Hy-Tracker framework comprises two key components. We introduce a hierarchical attention for band selection (HAS-BS) that selectively processes and groups the most informative spectral bands, thereby significantly improving detection accuracy. Additionally, we have developed a refined tracker that refines the initial detections by incorporating a classifier and a temporal network using gated recurrent units (GRUs). The classifier distinguishes similar objects, while the temporal network models temporal dependencies across frames for robust performance despite occlusions and scale variations (SVs). Experimental results on hyperspectral benchmark datasets demonstrate the effectiveness of Hy-Tracker in accurately tracking objects across frames and overcoming the challenges inherent in detection-based hyperspectral object tracking (HOT).**

***Index Terms*— Deep learning, hyperspectral tracker, object tracking, YOLO.**

Manuscript received 10 May 2024; revised 3 June 2024; accepted 15 June 2024. Date of publication 24 June 2024; date of current version 5 July 2024. *(Corresponding author: Jun Zhou.)*

Mohammad Aminul Islam is with the School of Information and Communication Technology, Griffith University, Nathan, QLD 4111, Australia, and also with the Department of Computer Science and Mathematics, Bangladesh Agricultural University, Mymensingh 2202, Bangladesh (e-mail: mohammadaminul.islam@griffithuni.edu.au).

Wangzhi Xing and Jun Zhou are with the School of Information and Communication Technology, Griffith University, Nathan, QLD 4111, Australia (e-mail: w.xing@griffith.edu.au; jun.zhou@griffith.edu.au).

Yongsheng Gao and Kuldip K. Paliwal are with the Institute for Integrated and Intelligent Systems, Griffith University, Nathan, QLD 4111, Australia (e-mail: yongsheng.gao@griffith.edu.au; k.paliwal@griffith.edu.au).

Digital Object Identifier 10.1109/TGRS.2024.3418337

I. Introduction

OBJECT tracking is one of the fundamental and ongoing research topics in computer vision and remote sensing and has many applications in traffic monitoring [1], autonomous vehicles [2], and video surveillance [3]. The primary objective of object tracking is to determine the size and location of a specific object and accurately trace its trajectory across consecutive video frames, relying on the initial position and size of the target object [4]. Traditional object trackers designed on RGB videos encounter difficulties in some challenging scenarios, including background clutters (BCs), similar color and texture of objects and their background, the close appearance of multiple objects in the same category, and significant deformation [5]. In contrast, hyperspectral videos offer a significant advantage in these scenarios as they capture a broader range of light wavelengths and higher spectral resolution compared to RGB videos [6]. Each pixel in hyperspectral videos contains detailed spectral information, enabling precise analysis and identification of materials and objects [7]. This distinctive spectral signature empowers the tracker to follow objects better [8], provide higher discrimination capabilities, and enhance object tracking accuracy in challenging scenarios [9].

Different researchers have proposed various methods for hyperspectral object tracking (HOT). Qian et al. [10], Henriques et al. [11], Xiong et al. [12], Chen et al. [13], and Hou et al. [14] utilized correlation filter-based methods for object tracking. However, these methods often rely on handcrafted features or simplistic feature representations, which are inherently limited to capturing linear relationships between features. This limitation may not be sufficient for modeling complex object motion and deformation. Moreover, these methods are susceptible to occlusions, scale variations (SVs), and the similarity of objects to their backgrounds.

Therefore, to extract discriminative features and intricate patterns from hyperspectral data, Uzkent et al. [15], Liu et al. [16], and Liu et al. [17] employed deep learning-based methods. However, the performance of these methods is often limited due to the insufficient training dataset. To mitigate this problem, Li et al. [18], [19], [22] and Islam et al. [20], [21] split the hyperspectral bands into multiple pseudo-color images and passed them through trackers, which produce

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:57 UTC from IEEE Xplore. Restrictions apply.

several weak trackers. However, it is noted that weak trackers can only partially extract the object's spectral information, limiting their ability to capture highly discriminative features.

To obtain better semantic information, Zhao et al. [8], [9], [23] integrate both RGB and hyperspectral data. The fusion of hyperspectral and RGB data in tracking enhances the capabilities of the tracking system by leveraging the strengths of both modalities. However, this fusion strategy is challenging when the data is collected from different viewpoints, leading to misalignments and further exacerbating the issues of occlusions, SVs, and similar object-background appearances.

The performance of the aforementioned deep learning-based hyperspectral trackers is constrained by several factors. A primary challenge is the scarce availability of labeled training data, which limits these models' ability to generalize effectively across different challenging scenarios. Furthermore, the high dimensionality and inherent complexity of hyperspectral data present significant obstacles in feature extraction and representation learning, thereby adversely affecting the trackers' efficiency. Additionally, a notable limitation is the absence of temporal flow integration within their frameworks. This omission results in an inability to capture and utilize dynamic information effectively over time. Therefore, there is a pressing need for a framework that can effectively handle data dependency issues and improve tracking accuracy by addressing real-world tracking challenges.

Despite the renowned object detection capabilities of YOLO [24] in RGB images, to the best of our knowledge, it has not been extensively applied in HOT. This gap is primarily due to the complexity of adapting YOLO to handle multiple spectral bands, occlusions, and other inherent challenges in hyperspectral environments. To address this gap, we introduce Hy-Tracker, a novel tracking framework specifically designed for hyperspectral data. Hy-Tracker employs a novel hierarchical attention for band selection (HAS-BS) network to effectively select and group the most informative spectral bands from hyperspectral images (HSIs), enhancing the detector's sensitivity to relevant features. Additionally, we have designed a refined tracker on top of YOLO's detections, utilizing a classifier and a sequential temporal network with gated recurrent units (GRUs). The classifier distinguishes similar objects, while the temporal network explores time-domain correlations between objects in different frames, ensuring robust tracking despite variations in appearance due to motion or illumination changes. Furthermore, to address data dependency issues and improve the tracker's robustness and generalization capabilities, we leverage a large RGB dataset augmented with a Gaussian cut-paste method. This approach ensures that Hy-Tracker remains adaptable to diverse hyperspectral environments while maintaining high accuracy and efficiency in challenging tracking scenarios.

Our contribution is four-folded, which is listed below.

1) A novel tracking framework, Hy-Tracker, is introduced, which leverages the strengths of YOLOv7 [25] for the very first time and addresses the unique challenges posed by hyperspectral tracking.
2) We propose a novel HAS-BS to select informative bands from HSIs effectively.
3) A refined tracking module, consisting of a classifier and a sequential temporal network, has been developed to improve the accuracy of initial detections and maintain the temporal correlations of the object across successive video frames.
4) The experimental results on hyperspectral benchmark datasets validate the robustness and effectiveness of Hy-Tracker in achieving accurate object tracking from hyperspectral videos.

The remaining of this article is organized as follows. In Section II, we conduct a comprehensive review of related works. Section III presents our proposed tracking framework, providing an in-depth discussion of its components and methodology. In Section IV, we present and discuss the experimental results, including the datasets, the evaluation metrics, and the performance of Hy-Tracker in various tracking scenarios. Finally, we draw conclusion in Section V.

II. Related Work

This section reviews existing literature about object tracking in hyperspectral imagery, focusing on different approaches, methodologies, and challenges encountered.

Hyperspectral trackers can be broadly categorized into two groups [22]: correlation filter (CF) methods and deep learning-based methods. Correlation filtering-based methods rely on correlation filters to establish a relationship between the object's features and the surrounding context in the hyperspectral video. Qian et al. [10] first combined convolutional networks with the kernelized correlation filter (KCF) [11] framework to track objects in hyperspectral videos. They used normalized 3-D cubes as fixed convolution filters to encode local spectral–spatial features and then applied KCF for tracking. Xiong et al. [12] extracted material information of the target object using the spectral–spatial histogram of multidimensional gradients (SSHMGs) as 3-D local spectral–spatial structures and fractional abundances of material components. Discriminative spatial–spectral features can also be extracted using spatial–spectral convolution kernels in the Fourier transform domain [26], histogram of oriented mosaic gradient descriptors [13], or adaptive spatial–spectral discriminant analysis [27]. Hou et al. [14] constructed a CF-based framework using a tensor sparse CF with a spatial–spectral weighted regularizer to reduce spectral differences in homogeneous backgrounds and to penalize filter templates based on spectral dissimilarity.

In contrast, deep learning-based methods extract discriminative features and intricate patterns directly from hyperspectral data, showcasing notable performance within HOT. Uzkent et al. [15] introduced an online generative hyperspectral tracking method that relies on likelihood maps to aid tracking without needing offline classifiers or extensive hyperparameter tuning. They further used KCF and deep convolutional neural network (CNN) features for aerial object tracking in the hyperspectral domain [28]. Liu et al. [16] introduced the HA-Net, employing an anchor-free Siamese network with a spectral classification branch. This approach

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:57 UTC from IEEE Xplore. Restrictions apply.

leverages all hyperspectral bands during training, significantly augmenting object identification capabilities. Subsequently, Liu et al. [17] presented the H3-Net, a framework designed for high spectral–spatial–temporal resolution hyperspectral tracking. This approach encompasses an unsupervised training strategy and a dual-branch Siamese network structure. The fusion of deep learning and discriminative correlation filters (DCFs) within the H3-Net framework further enhances feature compatibility and overall tracking performance.

While these deep learning-based trackers exhibit promising performance in the hyperspectral domain, their performance is limited by insufficient training datasets. To address this challenge, Li et al. [18] introduced BAE-Net through an encoder and decoder architecture, which splits the hyperspectral bands into multiple three-channel pseudo-color images based on band-wise weights generated by the band attention model. These pseudo-color images are fed into the adversarial learning-based tracker VITAL [29], producing several weaker trackers that are combined through ensemble learning to determine the precise object location. In another method, they [19] split the hyperspectral bands into multiple pseudo-color images based on the score of the spatial–spectral–temporal attention module. However, it is noted that the weak trackers in [18] and [19] partially extract the object spectral information and thus limit their ability to capture highly discriminative features. Therefore, SiamF [30] is introduced by fusing material information with the Siamese network. SiamF incorporates a hyperspectral feature fusion (HFF) module with dense connection architecture. This module integrates features from various layers and band groups, employing global–local channel attention to create a comprehensive spatial–spectral representation. Additionally, SiamF incorporates online spatial and material classifiers for adaptive online tracking. Tang et al. [31] introduced a heterogeneous encoder–decoder (HED) and spectral semantic representation (SSR) modules. These modules are designed for the extraction of spatial and spectral semantic features. They also adopted a two-stage training approach to learn the relevant parameters. The acquired spatial–spectral representations are subsequently merged to estimate the optimal target state.

Deep learning models demonstrate the capacity to acquire hierarchical data, enabling them to extract relevant features from the datasets autonomously. A substantial amount of data is required to learn these intricate features. Models trained on limited datasets often manifest overfitting, resulting in memorizing the training data. Furthermore, the models trained solely on hyperspectral data, neglecting the valuable information insights in RGB videos. To bridge the gap, Liu et al. [32] introduced SiamHYPER, a dual deep Siamese network framework, incorporating a pretrained RGB-based tracker and a hyperspectral target-aware module. Li et al. [22] developed a SiamBAG framework, where the model undergoes training using extensive RGB data instead of hyperspectral data. They initially employed a band attention module to group hyperspectral bands into multiple pseudo-color images. These images are then processed by SiamBAG, which combines classification responses from various branches to improve object localization. Lei et al. [33] introduced SSDT-Net by employing transfer learning to adapt knowledge from traditional color videos for hyperspectral tracking. It incorporates a dual-transfer strategy to evaluate the similarity between the source and target domains, optimizing the utilization of deep learning models. Tang et al. [34] designed a BAHT, which describes the semantic features of target appearances by utilizing pre-trained backbone networks that had been trained on color videos.

To obtain better semantic information, some researchers used both RGB and hyperspectral data [8], [9]. Zhao et al. [9] obtained modality-specific features from RGB and hyperspectral data using a dual-branch Siamese network and transformer. This method was extended to a dual-branch transformer tracking (RANet) architecture, guided by modality reliability to ensure that the most informative feature is used to improve tracking accuracy. However, data from different viewpoints may not be geometrically well aligned. In certain instances, some objects or areas of interest may be visible from one viewpoint but not from the other. This discrepancy makes it challenging to fuse information effectively.

Despite the advancements in object tracking techniques for hyperspectral videos, several challenges persist. These challenges include handling occlusions, SVs, and the limited availability of annotated datasets for training deep learning models. Additionally, adapting existing algorithms designed for RGB imagery to hyperspectral data poses inherent challenges due to the unique properties of hyperspectral imagery, such as multiple spectral bands and varying spectral signatures. In Section III, we present our proposed tracking framework, Hy-Tracker, which addresses these research gaps by leveraging state-of-the-art techniques and methodologies tailored specifically for HOT.

III. Hy-Tracker Framework

In this section, we will discuss the proposed Hy-Tracker framework. The Hy-Tracker framework divides a hyperspectral video frame (HVF) into numerous three-channel pseudo-color images based on the hierarchical attention scores. These pseudo-color images are then fed into the YOLOv7 for generating the candidate's proposal. After that, the candidates are refined to pinpoint the accurate object size and location.

The architecture of the proposed Hy-Tracker is shown in Fig. 1. As shown in the figure, this framework consists of three main modules: HAS-BS, YOLO, and a tracker. Each of these modules is described below.

A. Hierarchical Attention for Band Selection

HSIs consist of numerous bands, each containing unique information about the objects in the scene. However, not all bands contribute equally to object tracking, and processing all bands can lead to inefficiencies and increased computational costs. Additionally, variations in object sizes across different videos further complicate band selection, particularly in video tracking scenarios.

Conventional methodologies for hyperspectral band selection include selecting three fixed bands [35] or processing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:57 UTC from IEEE Xplore. Restrictions apply.

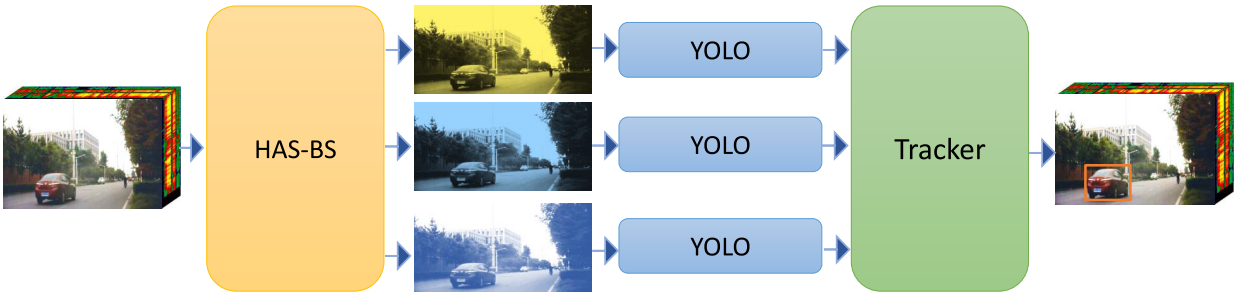


Fig. 1. This figure illustrates the architecture of Hy-Tracker, which comprises three main modules: HAS-BS, YOLO detection, and a tracker. At first, the HAS-BS technique splits the bands, selects the top nine bands based on the hierarchical attention scores, and then regroups them into three three-channel pseudo-color images. After that, candidate proposals are generated by sending the images to the YOLO. Finally, these proposals undergo a refinement process using an advanced tracker capable of handling challenges, including occlusions, FM, and BC.

the entire hyperspectral datacube [9]. However, fixed bands may overlook crucial information, while analyzing the entire datacube can introduce unnecessary complexity into the model [20]. Alternatively, dividing an HSI into multiple pseudo-color images or band groups, each containing a three-channel image, can serve as input for the model. However, the challenge lies in selecting the most informative bands. To address this, methods such as joint band entropy [36], contrast difference [37], and cross-band spectral–spatial similarity difference [20] have been employed. Nevertheless, these approaches rely on hand-crafted features and may not fully capture the inherent nature of spectral bands. Hence, deep learning-based attention mechanisms [18], [22], [30] have been introduced to learn dependencies among band groups.

However, traditional attention methods often prioritize final layer outputs, potentially overlooking valuable information present in lower levels of the data hierarchy, where lower levels capture fine-grained details while higher levels abstract more general features or patterns. Based on this intuition, we designed the HAS-BS network. HAS-BS selectively prioritizes and chooses the most informative bands from HSIs through a learned hierarchical attention mechanism from lower levels of data to higher levels of semantic information. This approach enhances tracking performance by focusing on bands that provide the most relevant information for distinguishing and tracking objects across varied conditions and scenes.

The HAS-BS architecture is shown in Fig. 2, which consists of four convolutional and four deconvolutional layers arranged hierarchically. Each convolutional layer captures features at a different level of abstraction, while the corresponding deconvolutional layers reconstruct the input from these features. Let us consider an HVF frame denoted as $\mathbf{B}$ with a total of m bands. The hierarchical feature extraction process can be represented as

$$\mathbf{F}_1 = \text{Conv}(\mathbf{B}) \tag{1}$$
$$\mathbf{F}_2 = \text{Conv}(\mathbf{F}_1) \tag{2}$$
$$\mathbf{F}_3 = \text{Conv}(\mathbf{F}_2) \tag{3}$$
$$\mathbf{F}_4 = \text{Conv}(\mathbf{F}_3) \tag{4}$$

where Conv() represents the convolution operation. These hierarchical features are then processed by four deconvolution layers to reconstruct the input from these features

$$\mathbf{D}_1 = \text{DeConv}(\mathbf{D}_2) + \text{Dropout}(\mathbf{F}_1) \tag{5}$$
$$\mathbf{D}_2 = \text{DeConv}(\mathbf{D}_3) + \text{Dropout}(\mathbf{F}_2) \tag{6}$$
$$\mathbf{D}_3 = \text{DeConv}(\mathbf{D}_4) + \text{Dropout}(\mathbf{F}_3) \tag{7}$$
$$\mathbf{D}_4 = \text{DeConv}(\mathbf{F}_4) \tag{8}$$

where, DeConv() represents the deconvolution operation and Dropout() represents the Dropout operation.

Subsequently, we calculate the total attention scores by employing five attention mechanisms in HAS-BS to select bands according to the hierarchical attention scores. The features from D_1 to D_4 and F_4 are passed to the attention mechanism to calculate the attention scores, which are defined by

$$\mathbf{W} = \sum_{i=1}^{5} A_i \tag{9}$$
$$\mathbf{A}_i = \text{Softmax}(\text{Linear}(\mathbf{F}_i)) \tag{10}$$

where linear represents linear operation, Softmax represents the softmax function along the channel dimension, and $\mathbf{F}_i$ represent features from $\mathbf{D}_1$ to $\mathbf{D}_4$ and $\mathbf{F}_4$.

Finally, the bands are ranked based on attention scores, resulting in three pseudo-color images. For example, if an HVF frame contains 16 bands, and the band order after ranking is $\{7, 10, 13, 15, 1, 5, 4, 14, 9, 12, 2, 6, 3, 8, 11, 16\}$, then the first nine bands $\{7, 10, 13, 15, 1, 5, 4, 14, 9\}$ are selected and reformed into three pseudo-color images $\{7, 10, 13\}$, $\{15, 1, 5\}$, and $\{4, 14, 9\}$. These images are then passed to YOLO for candidate generation.

B. YOLO

YOLO, known as "You Only Look Once," pioneered real-time object detection upon its introduction by Redmon et al. [24]. In our framework, we integrate YOLOv7 [25] as the object detectors. YOLOv7 introduces model re-parameterization to maintain performance during simplification, employs coarse-to-fine label assignment for enhanced object detection, and adopts a novel network topology to improve stability and reduce inference operations. Additionally, YOLOv7 leverages implicit knowledge transfer

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:57 UTC from IEEE Xplore. Restrictions apply.

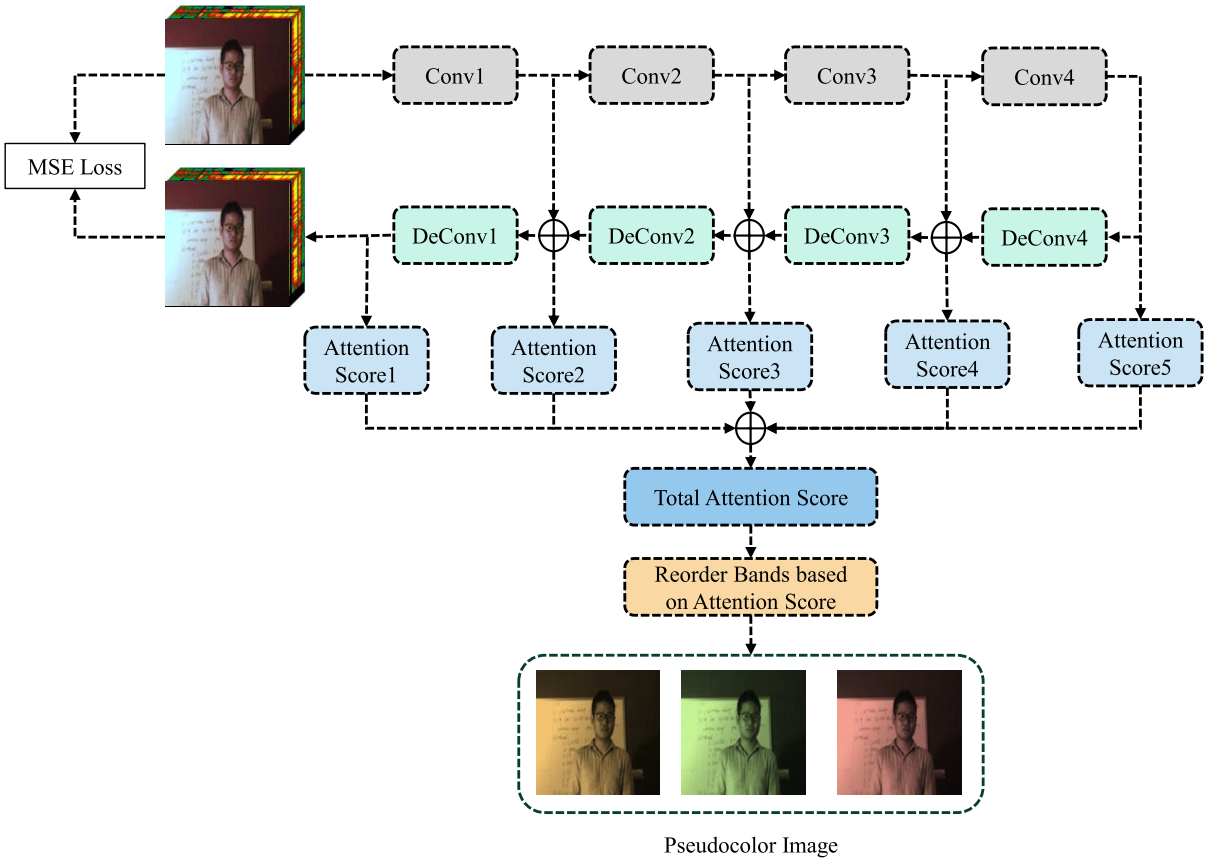


Fig. 2. Schematic architecture of the HAS-BS module. The HAS-BS model leverages convolutional layers for feature extraction, followed by deconvolutional layers for feature reconstruction. Five attention mechanisms are integrated at various levels of the network to determine the significance of each band hierarchically. Then, the bands are reordered according to the attention scores, and the top nine bands are selected and reformed in three pseudo-color images.

from pre-computed vectors and implements an exponential moving average (EMA) model to enhance model stability and performance.

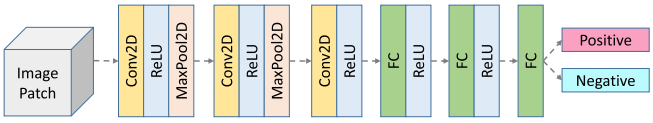


Fig. 3. Architecture of the classifier.

C. Trackers

The primary goal of YOLO is to detect all objects in a scene. In contrast, in our work, object tracking focuses on following a specific object of interest across multiple frames in a video sequence. The emphasis of the tracking is to maintain the identity and position of the object over time, which, however, is challenging in scenarios such as the similar appearance of multiple objects, BC, occlusions, and low spatial resolution. To address these challenges, a tracker is essential to bridge the detection and the requirement of object tracking. The tracker will transform the detection into a coherent and reliable tracking system by handling the complexity of the dynamic scenes. Therefore, we introduce the tracker which comprises three main modules: a classifier, a target proposal generator, and a sequence model for integrating the temporal flow.

1) Classifier: The purpose of the classifier is to separate the object of interest from other objects and backgrounds. The classifier architecture is presented in Fig. 3. It comprises three convolutional layers (conv1-3) and three fully connected layers (fc1-3) responsible for learning and extracting features from input. The reason for choosing a smaller network is that we are refining and fine-tuning the output of YOLO, and it is computationally efficient and suitable for online training.

2) Target Proposal Generator: The target proposal generator generates a set of potential bounding boxes that would likely include the target object within the current video frame. Inspired from [38], the potential candidates' bounding boxes are generated through Gaussian distribution sampling. This distribution is characterized by two main parameters: mean and covariance. The mean of the distribution is derived from the previous state of the target object, which estimates the potential position of this object within the current frame. The covariance matrix is diagonal and contains specific values $(0.09r^2, 0.09r^2, 0.25)$, where r represents the mean of the target object's bounding box width and height. In addition, the target object's size may be changed slightly between frames. Therefore, we adjust the scale of each candidate by multiplying with 1.05 sc where "sc" denotes the scaling value of the candidate box obtained from the Gaussian distribution.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:57 UTC from IEEE Xplore. Restrictions apply.

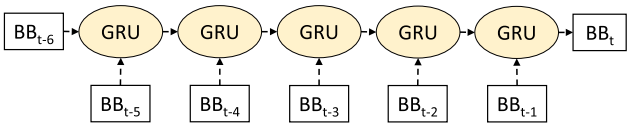


Fig. 4. Architecture of the temporal network consists of six GRU units. It predicts the future position of an object based on the sequence of its previous six bounding boxes.

3) Temporal Network: The temporal network employs GRUs [39] based architecture, specifically designed to predict the future positions of objects using sequential bounding boxes extracted from previous video frames. This architecture leverages the inherent capabilities of GRUs to effectively process temporal sequences, ensuring that long-term dependencies are maintained and utilized for accurate predictions. The schematic representation of our network is shown in Fig. 4, illustrating the flow of data through consecutive GRU layers.

The input to our model comprises six sequences of bounding boxes, each characterized by spatial coordinates and dimensions. These bounding boxes represent the position and size of an object in a frame and are denoted as BB_{t-1} through BB_{t-6}. Each bounding box is represented by (x, y, w, h), where x and y represent the coordinates of the top-left corner, and w and h indicate width and height, respectively.

The network consists of six GRU layers arranged sequentially. Each layer processes one bounding box at a time, along with the hidden state passed from the previous layer, which encapsulates information from earlier in the sequence. The operations within each GRU unit are defined by the following equations:

$$\mathbf{z}_t = \sigma\left(\mathbf{W}_z \cdot \left[\mathbf{h}_{t-1}, x_t\right] + \mathbf{b}_z\right) \tag{11}$$

$$\mathbf{r}_t = \sigma\left(\mathbf{W}_r \cdot \left[\mathbf{h}_{t-1}, x_t\right] + \mathbf{b}_r\right) \tag{12}$$

$$\tilde{\mathbf{h}}_t = \tanh\left(\mathbf{W} \cdot \left[\mathbf{r}_t \odot \mathbf{h}_{t-1}, x_t\right] + \mathbf{b}\right) \tag{13}$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t. \tag{14}$$

Here, $\mathbf{z}_t$, $\mathbf{r}_t$, $\tilde{\mathbf{h}}_t$, and $\mathbf{h}_t$ represent the update gate, reset gate, candidate gate, and final state, respectively, $\mathbf{W}_z$, $\mathbf{W}_r$, and $\mathbf{W}$ are weight matrices for the update gate, reset gate, and candidate state, respectively. $\mathbf{b}_z$, $\mathbf{b}_r$, and $\mathbf{b}$ are biased, and $[h_{t-1}, x_t]$ denotes the concatenation of the previous hidden state and the current input. The output of each GRU is passed to a fully connected layer, which maps the high-dimensional hidden state to the 4-D bounding boxes and passes to the next GRUs.

D. Offline Training and Online Tracking

1) Offline Training: The YOLO, HAS-BS, Classifier, and Temporal network are extensively trained on the HOT2020 dataset, where each video is labeled as a distinct object. However, the scarcity of annotated hyperspectral training data poses a formidable challenge during the training process. Therefore, in order to enhance the generalization capability and address the limited availability of data, we adopt pre-trained weights of YOLO, which is trained on the COCO dataset. This approach allowed the model to learn richer visual representations from a diverse and extensive dataset, thereby improving its ability to generalize to new, unseen data despite the scarcity of labeled samples in the training dataset.

In real-world scenarios, objects may not fit within the defined categories, and disparities between training and testing dataset objects abound. To address this issue, we use the initial frame of each video of the testing dataset. However, relying solely on a single frame might lead the model to overfit. To mitigate the overfitting, we implement the Gaussian cut-paste method to simulate object movement across the frame. The Gaussian cut-paste method leverages the properties of a 2-D Gaussian distribution to synthesize diverse instances of objects within the initial frame, mimicking their movement and appearance variations across the frame. This augmentation technique helps to mitigate the risk of overfitting while enhancing the model's ability to generalize to real-world scenarios where objects may exhibit variability in appearance and positioning. The probability distribution function for a 2-D Gaussian distribution with means (μ_p, μ_q) and standard deviations (σ_p, σ_q), given a specific object position (p, q), is expressed as follows:

$$f(p,q) = \frac{1}{2\pi\sigma_p\sigma_q} \exp\left(-\left(\frac{(p-\mu_p)^2}{2\sigma_p^2} + \frac{(q-\mu_q)^2}{2\sigma_q^2}\right)\right). \tag{15}$$

2) Online Tracking: The online tracking process begins with the generation of positive and negative candidate samples, denoted as C^+ and C^-, respectively, around the initial target state using the target proposal generation method outlined in Section III-C2. These candidates serve as training data to train the classifier, which distinguishes between positive and negative candidates.

During the tracking process, two types of predictions are generated for each frame: predictions from the YOLO model and predictions from the temporal network. Additionally, a set of target candidates denoted as N is derived from the previous target state.

While the YOLO model is proficient at predicting multiple objects within a frame, instances of similar background objects may lead to redundant detections. To address this, a distance threshold is applied to confine the YOLO predictions. Subsequently, both the target samples and YOLO's predictions are subjected to classification by a classifier, which assigns classification scores to each candidate. The optimal target state is then determined based on the candidate with the highest classification score.

However, in cases of occlusion, if a similar background object is present in the frame, the tracker may deviate from the intended target and follow the wrong object, as demonstrated in Fig. 5. Here, we observe that these deviations are noticeable in both the rider2 and playground videos. To address this challenge, we integrate the sequence model by incorporating a scale change threshold mechanism coupled with the temporal network, which accounts for the object's motion dynamics and predicts its future state based on past observations. If there exists a scale change of over 10% between the optimal target state and the previous state, the temporal network prediction

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:57 UTC from IEEE Xplore. Restrictions apply.

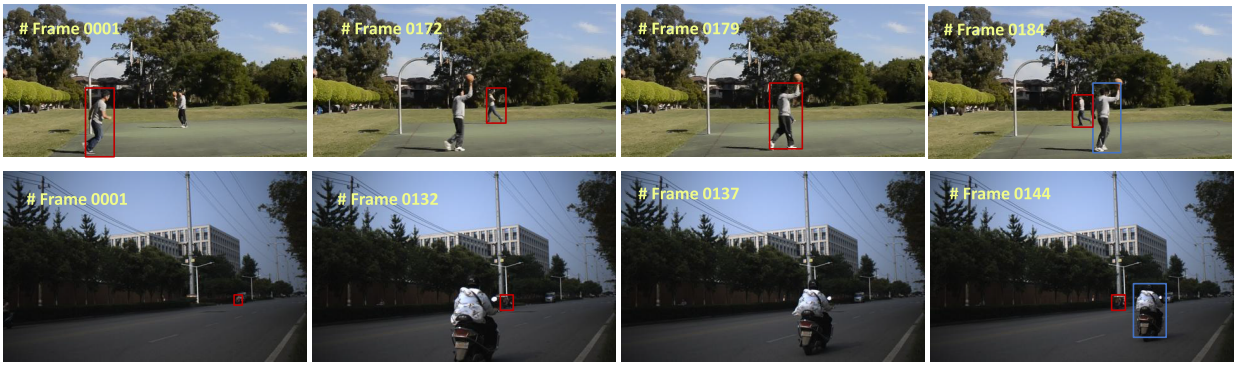


Fig. 5. Illustration of drifting away from the object of interest in both rider2 and playground videos, highlighting the challenges of occlusion and SV. The red trajectory denotes the actual object path, while the blue trajectory represents the tracker's estimated path.

Algorithm 1 Online Tracking Algorithm

Input: Initial target state $\tilde{x}_1$
Output: Predicted target state $\tilde{x}_t$

Generate positive candidates C^+ and negative candidates C^- around $\tilde{x}_1$ using III-C2.
Train the classifier using C^+ and C^-.
$\mathcal{T} \leftarrow \{1\}$, $\delta \leftarrow 25$.
for each frame in the video sequence **do**
  Generate target candidate $\tilde{x}_t^i$.
  $O_{y_n} \leftarrow$ YOLO's predictions $(ED(O_y^i, \tilde{x}_{t-1}) \leq \delta)$
  $O_k \leftarrow$Temporal Network Prediction
  $\tilde{x}_t \leftarrow$ optimal target state between $(\tilde{x}_t^i,\ O_{y_n})$
  if SC$(\tilde{x}_t, \tilde{x}_{t-1}) > 10\%$ **then**
    $\tilde{x}_t \leftarrow O_k$
    $\delta = \delta * 1.05$
  else
    $\delta = 25$
  end if
  Generate candidates C^+ and C^- around $\tilde{x}_t$
  $\mathcal{T} \leftarrow \mathcal{T} \cup \{t\}$
  if t mod 10 **then**
    Update (fcw1:fcw3) using $C^+_{v \epsilon \mathcal{T}}$ and $C^-_{v \epsilon \mathcal{T}}$
  end if
end for

* SC = Scale Change, ED = Euclidean Distance, and δ = Distance Threshold

is selected as the optimal target. The overall procedure of the tracking algorithm is presented in Algorithm 1.

IV. Experimental Results and Discussion

In this section, we provide details about the experimental setup, evaluation metrics, ablation studies, and a comparative analysis with state-of-the-art hyperspectral trackers.

A. Experimental Settings

In our study, we have employed the 2022 HOT challenges dataset.[1] This dataset was captured at a frame rate of 25 FPS using an XIMEA snapshot camera with an imec hyperspectral sensor. Each frame of the dataset was originally captured in a 2-D mosaic format, with wavelengths ranging from 470 to 620 nm. The dataset consists of a total of 40 training videos and 35 testing videos. It includes both RGB and HSI-False videos of the same scenes, where the HSI-False are the color videos converted from hyperspectral videos. Notably, the testing videos incorporate 11 challenging factors, which are deformation (DF), motion blur (MB), SV, out-of-view (OV), out-of-plane rotation (OPR), occlusion (OC), BC, in-plane rotation (IPR), fast motion (FM), illumination variation (IV), and low resolution (LR). The detailed description of the dataset is provided in [32].

All experiments were carried out on a machine featuring an Intel[2] Core i7-12700 CPU, 32 GB of RAM, and an NVIDIA RTX A4000 GPU equipped with 16 GB of dedicated graphics memory.

B. Learning Parameters

For training YOLO, we employed the SGD with an initial learning rate of 0.01, a final learning rate of 0.1 for the OneCycleLR scheduler, a momentum set to 0.937, a weight decay of 0.0005 for regularization, and a warm-up strategy spanning 3.0 epochs with an initial momentum of 0.8 and an initial bias learning rate of 0.1.

For training the classifier, we also used SGD with initial learning rates of 0.0001 for conv2d layers and 0.001 for fc layers. We used binary cross-entropy (BCE) as our loss function and set the momentum to 0.9 and weight decay to 0.0005.

The HAS-BS and Temporal Network were trained by using Adam optimizer with initial learning rates of 0.0001. We used MSE loss as the loss function for both networks.

C. Evaluation Metrices

In our experiments, four evaluation metrics are used, including success plots, precision plots, area under the curve (AUC), and precision rate (PR) at a given 20-pixel threshold (DP@20pixels). By following [25], we also included mAP@50, mAP@75, and mAP@90 as our evaluation metrics. It is worth noting that the one-pass evaluation (OPE) [12] technique was used to record all of the experimental results.

[1]https://www.hsitracking.com/

[2]Registered trademark.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:57 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF HY-TRACKER WITH DIFFERENT STATE-OF-THE-ARTS TRACKERS IN THE HOT-2022 DATASET. THE TOP TWO RESULTS ARE MARKED BY RED AND BLUE

	AUC	DP@20pixels	mAP@50	mAP@75	mAP@90	FPS
MHT [12]	0.586	0.882	0.876	0.554	0.274	2
SiamRPN++ [40]	0.645	0.920	0.934	0.685	0.336	19
MDNET [38]	0.610	0.936	0.861	0.566	0.268	5.2
BAE-Net [18]	0.606	0.878	0.874	0.667	0.333	<1
TransT [41]	0.649	0.910	0.937	0.741	0.441	25
CBFF-Net [2]	0.691	0.927	-	-	-	24
SST-Net [19]	0.623	0.916	0.884	0.646	0.326	<1
MFI [30]	0.601	0.893	0.831	0.583	0.293	2
SiamBAG [22]	0.632	0.892	0.899	0.701	0.365	5.7
SiamHYPER [32]	0.678	0.945	-	-	-	19
BABS [20]	0.616	0.901	0.913	0.672	0.341	4.23
VITAL [29]	0.604	0.897	0.842	0.570	0.287	5.27
Hy-Tracker	0.728	0.972	0.977	0.843	0.544	4.4

The success plot, where the x-axis is the overlap threshold and the y-axis is the success rate (SR), shows how well a tracker performs. The SR is determined by calculating the intersection over union (IoU) between the actual ground truth and the predicted bounding box. This measure is expressed as follows:

$$\mathrm{SR}(\tau) = \frac{N_{\mathrm{IoU}} > \tau}{N_{\mathrm{total}}}. \tag{16}$$

Here, τ represents a threshold value that ranges from 0 to 1, while "N" denotes the number of frames. If we denote A_p as the area of the predicted representation of the target object and A_g as the area of the ground truth representation, the IoU is defined as

$$\mathrm{IoU} = \frac{|A_g \cap A_p|}{|A_g \cup A_p|}. \tag{17}$$

Here, $\cap$ and $\cup$ indicate the IoU between the target object's ground truth and the predicted bounding boxes. A frame is considered to have been tracked successfully when the IoU exceeds the threshold τ. The proximity of the curve to the upper right corner of the plot signifies superior tracker performance.

The precision plot, where the x-axis represents the center position error (CLE) and the y-axis corresponds to the PR, provides a visual representation of a tracker's precision. The CLE is calculated as the Euclidean distance between the centers of the actual and predicted bounding boxes and is defined as follows:

$$\mathrm{CLE} = \sqrt{(p_1 - p_2)^2 + (q_1 - q_2)^2} \tag{18}$$

where $(p_1,\ q_1)$ and $(p_2,\ q_2)$ represent the center locations of the ground truth and predicted bounding boxes of the target object. The CLE difference between the ground truth and the predicted bounding box is measured as PR, which is defined as

$$PR(\tau) = \frac{N_{\mathrm{CLE}} > \eta}{N_{\mathrm{total}}} \tag{19}$$

where η is a threshold ranging from 0 to 50 pixels and N represents the number of frames. A frame is considered to have been tracked successfully if the CLE is less than η. The proximity of the curve to the upper left corner of the plot signifies superior tracker performance.

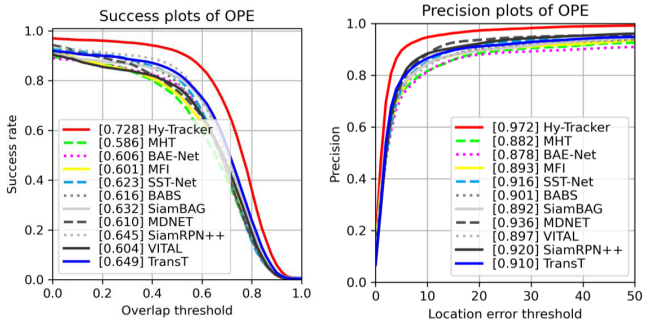


Fig. 6. Comparison between different hyperspectral trackers in terms of success plot and precision plot on the HOT-2022 dataset.

D. Performance Comparison on HOT-2022 Dataset

In this section, we conduct a comprehensive comparative analysis of Hy-Tracker in relation to 12 state-of-the-art tracking methods, including MHT [12], SiamRPN++ [40], SiamHYPER [32], MDNET [38], BAE-Net [18], TransT [41], SST-Net [19], VITAL [29], CBFF-Net [2], MFI [30], BABS [20], and SiamBAG [22].

1) Quantitative Comparison: The detailed quantitative results are presented in Table I. From these results, we observe that the Hy-Tracker achieves the highest scores in all evaluation metrics. The underlying reason is that MHT [12] and CBFF-Net [2] rely solely on hand-crafted features, while BAE-Net [18], BABS [20], and SST-Net [19] leverage hyperspectral information effectively by grouping bands into pseudo-color images and then producing several weak trackers. On the other hand, TransT [41], SiamHYPER [32], and SiamBAG [22] benefit from extensive color object tracking data during training, resulting in respectable AUC values of 0.649, 0.678, and 0.632, respectively. However, these models tend to overlook challenges such as occlusion, BC, and SV. In contrast, Hy-Tracker's refined tracking module adeptly addresses these challenges, thus ensuring remarkable accuracy and precision, rendering it robust for real-world scenarios. The visual representation in terms of the success and precision plots is depicted in Fig. 6.

2) Attributed Comparison: We conducted a comprehensive analysis comparing Hy-Tracker with other state-of-the-art trackers across various attributes of the dataset. The results of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:57 UTC from IEEE Xplore. Restrictions apply.

TABLE II

COMPARISON OF DIFFERENT HYPERSPECTRAL TRACKERS IN TERMS OF ATTRIBUTE ANALYSIS OF DIFFERENT CHALLENGING SCENARIOS IN THE HOT-2022 DATASET. THE TOP RESULTS ARE MARKED BY RED AND BLUE. EVALUATION IS PERFORMED USING THE AUC METRIC

	SV	OV	OPR	OCC	BC	DFE	FM	IPR	IV	LR	MB
MHT [12]	0.574	0.621	0.644	0.564	0.606	0.664	0.542	0.670	0.477	0.476	0.560
SiamRPN++ [40]	0.649	0.637	0.719	0.612	0.661	0.702	0.666	0.715	0.560	0.577	0.709
MDNET [38]	0.602	0.559	0.703	0.572	0.663	0.694	0.570	0.698	0.494	0.423	0.585
BAE-Net [18]	0.604	0.509	0.701	0.554	0.651	0.680	0.608	0.699	0.524	0.486	0.593
TransT [41]	0.660	0.787	0.741	0.610	0.652	0.722	0.662	0.724	0.582	0.561	0.662
CBFF-Net [2]	0.668	0.694	0.739	0.629	0.705	0.758	0.703	0.740	0.580	0.657	0.773
SST-Net [19]	0.602	0.473	0.699	0.594	0.685	0.700	0.561	0.696	0.501	0.462	0.536
MFI [30]	0.599	0.607	0.680	0.546	0.651	0.639	0.600	0.692	0.515	0.514	0.570
SiamBAG [22]	0.622	0.622	0.683	0.596	0.648	0.692	0.615	0.703	0.533	0.583	0.650
SiamHYPER [32]	0.657	0.596	0.706	0.635	0.702	0.720	0.710	0.720	0.592	0.664	0.750
BABS [20]	0.600	0.588	0.679	0.592	0.641	0.700	0.559	0.678	0.584	0.495	0.538
VITAL [29]	0.592	0.610	0.707	0.576	0.641	0.669	0.632	0.708	0.494	0.364	0.613
Hy-Tracker	0.720	0.710	0.770	0.710	0.741	0.743	0.725	0.773	0.684	0.642	0.740

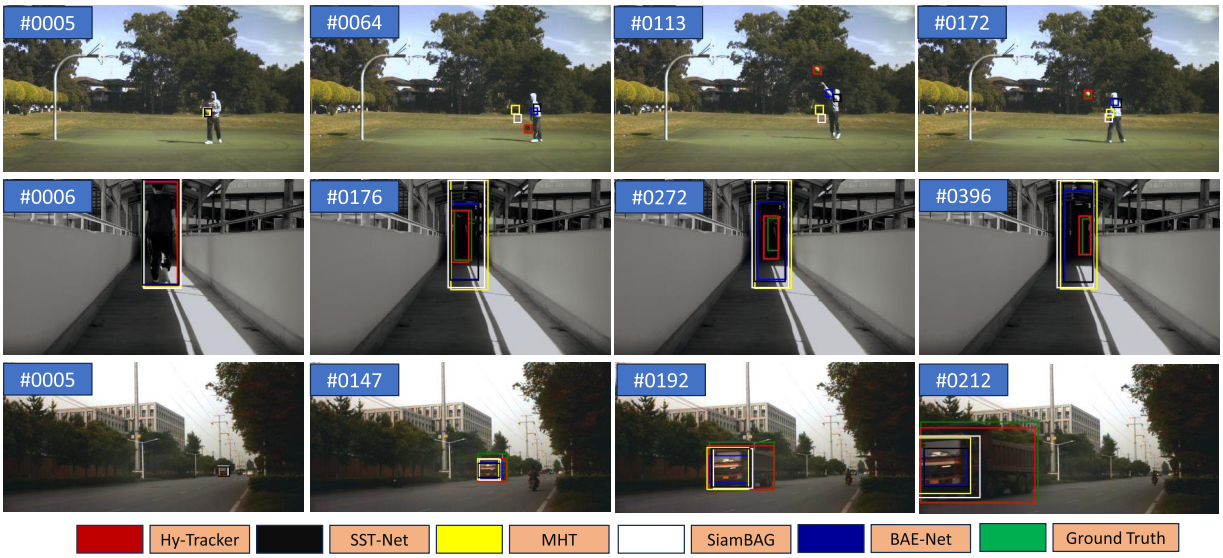


Fig. 7. Demonstration of some tracking results from basketball, student, and truck videos from the HOT-2022 dataset.

this comparison are summarized in Table II. Across multiple challenging scenarios, Hy-Tracker consistently outperformed all other trackers, achieving the highest scores in seven attributes: BC, FM, IPR, IV, occlusion (OCC), OPR, and SV.

Specifically, Hy-Tracker demonstrated superior performance in handling challenges such as BC, FM, IV, occlusion, and various forms of rotation and SV. It also achieved the second-highest score in the deformation (DFE) attribute, further highlighting its effectiveness across a wide range of scenarios.

While Hy-Tracker dominated in most attributes, CBFF-Net [2], TransT [41], and SiamHYPER [32] showcased notable performance in the LR and motion blur (MB) scenarios, respectively. CBFF-Net [2] attained the highest score in the MB and DFE scenarios, while SiamHYPER [32] and TransT [41] achieved the highest score in the LR and OV scenarios, respectively. These findings suggest that there are opportunities for further improvement in the future.

3) Qualitative Comparison: We also conducted a qualitative analysis of Hy-Tracker with four other state-of-art methods, including SST-Net [19], MHT [12], SiamBAG [22] and BAE-Net [18]. The qualitative result is depicted in Fig. 7. Upon thorough examination, Hy-Tracker consistently outperformed every other method across all video scenarios. Its robust performance was particularly evident in handling complex challenges such as MB, occlusion, and LR in the basketball video. Additionally, in scenarios involving player movements with rotation, deformation, and SV, Hy-Tracker demonstrated superior adaptability and accuracy compared to the other methods. Even in track video, which posed unique challenges like occlusion and IV, Hy-Tracker excelled in maintaining tracking accuracy and robustness, ensuring objects remained reliably tracked despite varying environmental conditions.

E. Performance Comparison on HOT-2023 Dataset

The performance of the Hy-Tracker is further evaluated on the HOT-2023 dataset, which consists of three different modules of data: VIR, RedNIR, and NIR, containing 16, 15, and 25 bands, respectively. This dataset includes 110 training videos and 87 validation videos. Additionally, it incorporates eight more challenging attributes, including camera motion (CM), low contrast (LC), reflections (RE), small target (ST), turning light on and off (TOO), low light (LL), red-NIR low

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:57 UTC from IEEE Xplore. Restrictions apply.

TABLE III
COMPARISON OF HY-TRACKER WITH DIFFERENT STATE-OF-THE-ARTS TRACKERS IN THE HOT-2023 DATASET. THE TOP TWO RESULTS ARE MARKED BY RED AND BLUE

	AUC	DP@20Pixels	mAP@50	mAP@75	mAP@90
SiamBAN [42]	0.531	0.756	0.770	0.488	0.246
SiamCAR [43]	0.554	0.768	0.790	0.555	0.246
SiamGAT [44]	0.561	0.770	0.815	0.598	0.296
TransT [45]	0.566	0.777	0.818	0.641	0.393
Hy-Tracker	0.642	0.847	0.881	0.726	0.482

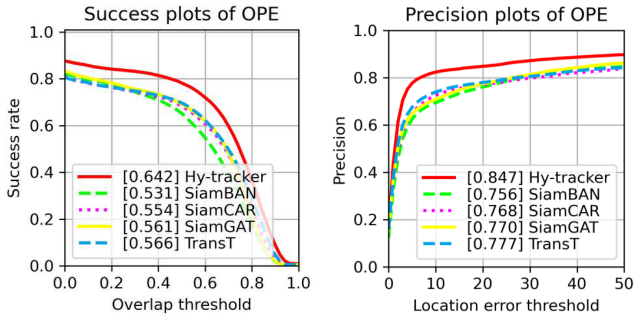


Fig. 8. Comparison between different hyperspectral trackers in terms of success plot and precision plot on the HOT-2023 dataset.

visibility (RLV), and scale change (SC). We compare the proposed Hy-Tracker method with four other state-of-the-art methods: SiamBAN [42], SiamGAT [44], SiamCAR [43], and TranST [45].

1) Quantitative Analysis: The quantitative comparison of Hy-Tracker with the other four state-of-the-art methods is depicted in Table III and the visual representation of the success plot and the precision plot is depicted in Fig. 8. Among the evaluated trackers, Hy-Tracker demonstrates notable performance, achieving competitive scores across all metrics. In terms of AUC and DP@20 pixels, Hy-Tracker stands out with scores of 0.642 and 0.847, respectively, surpassing the second-best performer, TransT [45], which achieves 0.566 and 0.777, respectively. Additionally, Hy-Tracker excels in other evaluation metrics, including mAP@50, mAP@75, and mAP@90, where it outperforms all other trackers, underscoring its effectiveness in accurately tracking objects in challenging scenarios. These results showcase its capability to achieve superior performance in real-world applications.

2) Attributed Comparison: The HOT-2023 dataset consists of 19 attributes, each defining a particular challenge scenario encountered in object tracking. The performance of Hy-Tracker relative to other state-of-the-art trackers is shown in Table IV. From this table, it becomes evident that Hy-Tracker consistently outperforms other trackers across numerous challenging scenarios. Notably, in instances involving BC, CM, FM, IV, IPR, LC, LR, MB, occlusion (OCC), OPR, OV, and SV, Hy-Tracker excels, demonstrating superior tracking accuracy and robustness.

However, Hy-Tracker outperforms by SiamBAN [42] in three attributes: LL, reflections (RE), and ST. This suggests that there are opportunities for further improvement in the future.

3) Qualitative Comparison: The qualitative analysis of Hy-Tracker for the three videos, vis-card19, vis-rainystreet16, and vis-playground, is depicted in Fig. 9. This analysis

TABLE IV
COMPARISON OF DIFFERENT HYPERSPECTRAL TRACKERS IN TERMS OF ATTRIBUTE ANALYSIS OF DIFFERENT CHALLENGING SCENARIOS IN THE HOT-2022 DATASET. THE TOP RESULTS ARE MARKED BY RED AND BLUE. EVALUATION IS PERFORMED USING THE AUC METRIC

	SiamBAN [42]	SiamCAR [43]	SiamGAT [44]	TransT [45]	Hy-Tracker
BC	0.474	0.521	0.485	0.493	0.645
CM	0.482	0.517	0.512	0.559	0.628
DFE	0.663	0.717	0.688	0.723	0.728
FM	0.608	0.66	0.651	0.653	0.681
IV	0.393	0.397	0.452	0.429	0.547
IPR	0.648	0.675	0.626	0.724	0.767
LC	0.126	0.18	0.177	0.288	0.390
LL	0.138	0.09	0.105	0.106	0.110
LR	0.482	0.506	0.52	0.501	0.614
MB	0.669	0.663	0.686	0.678	0.729
OCC	0.467	0.489	0.522	0.511	0.595
OPR	0.616	0.639	0.606	0.666	0.729
OV	0.292	0.33	0.316	0.446	0.529
RLV	0.423	0.333	0.548	0.244	0.607
RE	0.725	0.712	0.701	0.621	0.711
SC	0.719	0.677	0.729	0.664	0.735
SV	0.589	0.617	0.614	0.638	0.689
ST	0.66	0.608	0.625	0.524	0.596
TOO	0.273	0.223	0.366	0.281	0.396

provides a visual representation of Hy-Tracker's performance across various challenging scenarios, showcasing its ability to effectively track objects amidst occlusion, LC, camera movement, OPRs, and SV. Notably, in the vis-playground video, while other trackers may erroneously follow the wrong person after occlusion, due to the refined tracking module, Hy-Tracker demonstrates its capability to consistently follow the correct object, showcasing its ability to handle occlusion. Moreover, in the vis-rainystreet16 video, the tracker maintains its tracking accuracy even when the camera is in motion after frame 80, indicating its robustness in dynamic environments. Therefore, these results highlight the effectiveness of Hy-Tracker in diverse real-world scenarios.

F. Ablation Study

1) Contribution of Individual Modules in the Framework: In this section, we examine the individual contributions of YOLO, tracker, and temporal network into the framework. The result of this analysis is presented in Table V. It is worth mentioning that we choose the highest confidence score among all the candidate proposals when we consider only YOLO for tracking. From this table, we observe that using only YOLO, we achieve an AUC of 0.639, a DP@20pixels of 0.848, and a mAP@50 of 0.955. When we incorporate the temporal network, the AUC improves to 0.674, and the DP@20 pixels

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:57 UTC from IEEE Xplore. Restrictions apply.

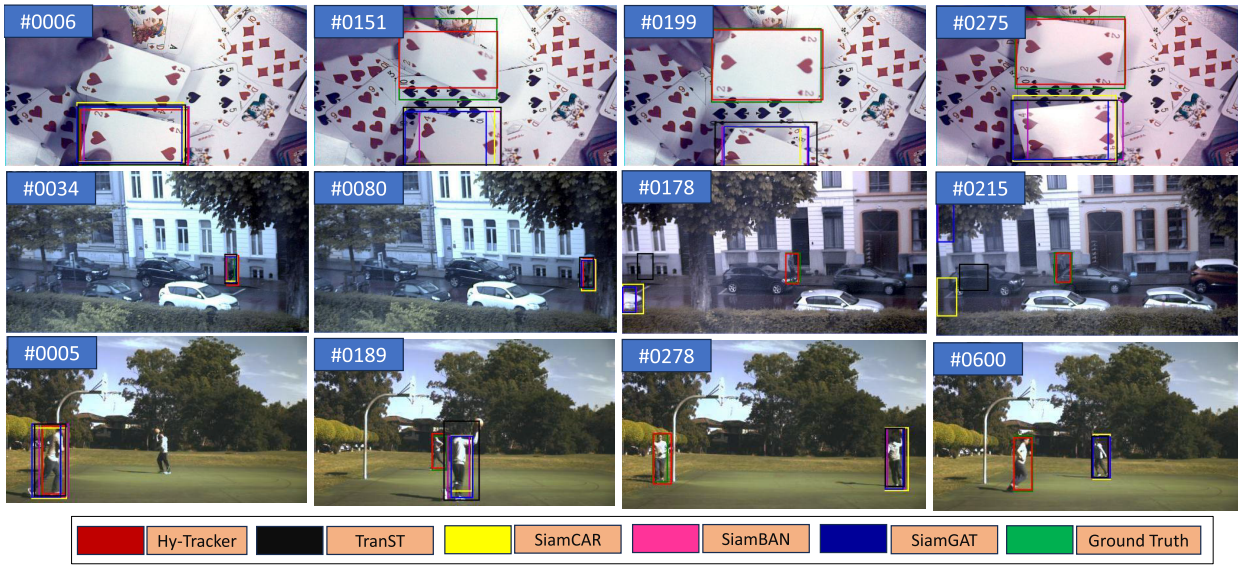


Fig. 9. Demonstration of some tracking results from vis-card19, vis-rainystreet16, and vis-playground videos from the HOT-2023 dataset.

TABLE V

ABLATION STUDY: CONTRIBUTION OF YOLO, TRACKER, AND TEMPORAL NETWORK INTO THE FRAMEWORK. THE BEST RESULT IS HIGHLIGHTED BY RED

YOLO	Tracker	Temporal Network	AUC	DP@20pixels	mAP@50
√			0.639	0.848	0.955
√		√	0.674	0.907	0.922
√	√		0.710	0.950	0.963
√	√	√	0.728	0.972	0.977

TABLE VI

ABLATION STUDY: COMPARISON WITH EXISTING BAND SELECTION TECHNIQUES. THE BEST RESULT IS HIGHLIGHTED BY RED

	AUC	DP@20pixels	mAP@50
SiamOHOT [35]	0.682	0.919	0.939
BABS [20]	0.707	0.955	0.959
BAE-Net [18]	0.692	0.933	0.945
HAS-BS (Hy-Tracker)	0.728	0.972	0.977

TABLE VII

ABLATION STUDY: EFFECT OF INCREMENTAL INCLUSION OF ATTENTION LAYERS ON MODEL PERFORMANCE. THE BEST RESULT IS HIGHLIGHTED BY RED

	AUC	DP@20pixels	mAP@50
Attention 1	0.691	0.926	0.943
Attention 1, 2	0.694	0.932	0.949
Attention 1, 2, 3	0.703	0.948	0.964
Attention 1, 2, 3, 4	0.714	0.962	0.971
Attention 1, 2, 3, 4, 5 (HAS-BS)	0.728	0.972	0.977

TABLE VIII

ABLATION STUDY: COMPARISON ON VIDEO SEQUENCE LENGTH. THE BEST RESULT IS HIGHLIGHTED BY RED

Sequence Length	AUC	DP@20pixels	mAP@50
4	0.688	0.928	0.938
5	0.698	0.935	0.951
6	0.728	0.972	0.977
7	0.703	0.946	0.963
8	0.699	0.938	0.956

TABLE IX

ABLATION STUDY OF IMPROVING CHALLENGING SCENARIOS (BC, OCC, IV, SV) WITH RESULTS EVALUATED USING AUC. THE BEST RESULT IS HIGHLIGHTED BY RED

	BC	OCC	IV	SV
YOLO	0.660	0.623	0.564	0.617
YOLO + Tracker	0.732	0.686	0.667	0.696
YOLO + Tracker + Temporal Network	0.741	0.710	0.684	0.720

increase to 0.907, suggesting that adding temporal information helps refine the localization of detections. Including the tracker with YOLO improves the AUC to 0.710, the DP@20pixels to 0.950, and the mAP@50 to 0.963, demonstrating that the tracker significantly enhances the robustness and consistency of detections. However, the most notable improvement is observed when we combine all of these components, with an AUC of 0.728, DP@20pixels of 0.972, and mAP@50 of 0.977, showing that the integrated approach leverages the strengths of each component effectively to achieve highly precise and accurate tracking and detection across frames.

2) Comparison With Existing Band Selection Methods: We compare the proposed HAS-BS of Hy-Tracker with three other HOT band selection methods, including SiamOHOT [35], BABS [20], and BAE-Net [18]. The comparison results are shown in Table VI. The HAS-BS approach outperforms the other methods in all evaluated metrics. Specifically, it achieves an AUC of 0.728, which is higher than SiamOHOT's 0.682, BABS's 0.707, and BAE-Net's 0.692. For the metric DP@20pixels, HAS-BS reaches 0.972, surpassing SiamOHOT's 0.919, BABS's 0.955, and BAE-Net's 0.933.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:57 UTC from IEEE Xplore. Restrictions apply.

TABLE X
ABLATION STUDY OF THE IMPACT OF DATA SCARCITY ON MODEL PERFORMANCE. THE BEST RESULT IS HIGHLIGHTED BY RED

	AUC	DP@20pixels	mAP@50
HOT-Dataset	0.629	0.835	0.949
HOT-Dataset + Pretrained Weight	0.653	0.884	0.908
HOT-Dataset + Gaussian Cut Paste	0.673	0.905	0.927
HOT-Dataset + Pretrained Weight+ Gaussian Cut Paste	0.728	0.972	0.977

In terms of mAP@50, HAS-BS also leads with 0.977, compared to 0.939 for SiamOHOT, 0.959 for BABS, and 0.945 for BAE-Net.

This performance enhancement can be attributed to the methodological differences among the approaches. SiamOHOT only considers fixed band numbers for all videos, which may not adapt well to varying conditions and targets across different scenes. BABS focuses on the spectral and spatial differences between the target object and its local neighborhood, which enhances the discriminative capability but may not fully capture the hierarchical structures and dependencies in the data. BAE-Net applies attention modules at the final outputs, which improve focus on relevant features but might miss earlier cues that are also important.

On the other hand, HAS-BS utilizes hierarchical attention modules that process attention from lower levels of data to higher levels of semantic information. This approach ensures that the model not only focuses on immediate, salient features but also incorporates broader contextual data, leading to more robust and accurate tracking across diverse hyperspectral video sequences.

3) Effect of Incremental Inclusion of Attention Layers on Model Performance: We examine the performance impact of incrementally including attention layers within the HAS-BS network, as detailed in Table VII. From this table, we observed that the first attention layer, which is computed on the final output of DeConvolutional layers, achieves an AUC of 0.691. Adding each subsequent attention layer progressively improves these metrics. This final configuration, which represents the full capability of HAS-BS, achieves the best results with an AUC of 0.728, DP@20pixels of 0.972, and mAP@50 of 0.977, demonstrating that each additional layer contributes to a better understanding of the data, thereby significantly enhancing tracking accuracy. This incremental and systematic addition of attention layers validates our approach of employing a comprehensive attention mechanism to improve the performance of hyperspectral tracking tasks significantly.

4) Comparison on Video Sequence Length: We evaluate the performance of the Hy-Tracker across different video sequence lengths to determine the optimal sequence length for the temporal network. The results of this comparison are shown in Table VIII. The performance metrics, including AUC, DP@20pixels, and mAP@50, indicate that the optimal performance is observed at a sequence length of 6, where the Hy-Tracker achieves an AUC of 0.728, DP@20pixels of 0.972, and mAP@50 of 0.977. For sequence lengths shorter or longer than 6, the performance tends to decline. Increasing the sequence length may overly burden the network by requiring it to learn more complex temporal dynamics, while decreasing the sequence length could underfit the model, preventing it from accurately predicting the position of objects. Therefore, we choose the sequence length of 6, which represents a balanced trade-off between having sufficient temporal context for accurate tracking and avoiding the inclusion of extraneous information that could complicate the model's tracking ability. This balance ensures that the Hy-Tracker can effectively leverage temporal information without being overwhelmed by it.

5) Impact of Individual Modules on Addressing Tracking Challenges: Additionally, we evaluate the impact of integrating these modules to address the challenges of occlusion (OCC), BC, IV, and SV. The result of this analysis is shown in Table IX. As illustrated, the incorporation of YOLO, Tracker, and temporal attention progressively enhances the performance across these challenging scenarios. Specifically, from YOLO to the complete integration of all modules, there are consistent improvements observed in AUC scores for each challenging condition, underscoring the effectiveness of Hy-Tracker in mitigating various tracking challenges.

6) Impact of Data Scarcity on Model Performance: We also investigate the influence of data scarcity on model performance by conducting a study on three key aspects: utilizing solely the HOT-2022 dataset, incorporating pre-trained weights of YOLO trained on the extensive COCO dataset, and employing Gaussian cut-paste data augmentation technique. The result of this analysis is presented in Table X. From the table, it is evident that the integration of the RGB dataset and the Gaussian cut-paste method significantly impacts the model performance across all evaluated metrics, where the AUC increased to 0.728 from 0.629 in HOT-Dataset. Additionally, the DP@20pixels and mAP@50 metrics also improve substantially. These findings highlight the efficacy of combination to mitigate the effects of data scarcity and improve the robustness and accuracy of object tracking models.

V. CONCLUSION

In this research, a novel hyperspectral tracker, Hy-Tracker, is introduced, which bridges the gap between state-of-the-art object detection methods and hyperspectral tracking. The tracker module in Hy-Tracker refines YOLO's proposals and addresses the challenges of dynamic scenes. The integrated temporal network helps the tracker overcome the challenges of occlusion and SV. The promising results of Hy-Tracker in both the HOT-2022 and HOT-2023 datasets imply the robustness and reliability of the proposed framework. For future work, we aim to optimize and adapt Hy-Tracker for deployment on low-power, low-resource embedded systems. This will involve enhancing the algorithm to run efficiently on embedded hardware, which is crucial for real-time applications where processing power and energy consumption are limited.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:57 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] D. Cerutti-Maori, I. Sikaneta, and C. H. Gierull, "Optimum SAR/GMTI processing and its application to the radar satellite RADARSAT-2 for traffic monitoring," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3868–3881, Oct. 2012.

[2] L. Gao et al., "CBFF-Net: A new framework for efficient and accurate hyperspectral object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506114.

[3] J. Shao, B. Du, C. Wu, and L. Zhang, "Tracking objects from satellite videos: A velocity feature based correlation filter," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7860–7871, Oct. 2019.

[4] J. Yang, Z. Pan, Z. Wang, B. Lei, and Y. Hu, "SiamMDM: An adaptive fusion network with dynamic template for real-time satellite video single object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5608619.

[5] Y. Su, S. Mei, G. Zhang, Y. Wang, M. He, and Q. Du, "Gaussian information entropy based band reduction for unsupervised hyperspectral video tracking," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 791–794.

[6] Y. Wang, Y. Liu, G. Zhang, Y. Su, S. Zhang, and S. Mei, "Spectral–spatial-aware transformer fusion network for hyperspectral object tracking," in *Proc. 12th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Sep. 2022, pp. 1–5.

[7] C. Yu, R. Han, M. Song, C. Liu, and C.-I. Chang, "Feedback attention-based dense CNN for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2021.

[8] C. Zhao, H. Liu, N. Su, L. Wang, and Y. Yan, "RANet: A reliability-guided aggregation network for hyperspectral and RGB fusion tracking," *Remote Sens.*, vol. 14, no. 12, p. 2765, Jun. 2022.

[9] C. Zhao, H. Liu, N. Su, and Y. Yan, "TFTN: A transformer-based fusion tracking framework of hyperspectral and RGB," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5542515.

[10] K. Qian, J. Zhou, F. Xiong, H. Zhou, and J. Du, "Object tracking in hyperspectral videos with convolutional features and kernelized correlation filter," in *Smart Multimedia. ICSM 2018* (Lecture Notes in Computer Science), Toulon, France. Cham, Switzerland: Springer, 2018, pp. 308–319.

[11] J. F. Henriques, R. Caseiro, P. Martins, and J. Batista, "High-speed tracking with kernelized correlation filters," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 3, pp. 583–596, Mar. 2015.

[12] F. Xiong, J. Zhou, and Y. Qian, "Material based object tracking in hyperspectral videos," *IEEE Trans. Image Process.*, vol. 29, pp. 3719–3733, 2020.

[13] L. Chen, Y. Zhao, J. C.-W. Chan, and S. G. Kong, "Histograms of oriented mosaic gradients for snapshot spectral image description," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 79–93, Jan. 2022.

[14] Z. Hou, W. Li, J. Zhou, and R. Tao, "Spatial–spectral weighted and regularized tensor sparse correlation filter for object tracking in hyperspectral videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541012.

[15] B. Uzkent, A. Rangnekar, and M. J. Hoffman, "Aerial vehicle tracking by adaptive fusion of hyperspectral likelihood maps," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 233–242.

[16] Z. Liu et al., "An anchor-free Siamese target tracking network for hyperspectral video," in *Proc. 11th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Mar. 2021, pp. 1–5.

[17] Z. Liu, Y. Zhong, X. Wang, M. Shu, and L. Zhang, "Unsupervised deep hyperspectral video target tracking and high spectral–spatial–temporal resolution (H^3) benchmark dataset," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5513814.

[18] Z. Li, F. Xiong, J. Zhou, J. Wang, J. Lu, and Y. Qian, "BAE-Net: A band attention aware ensemble network for hyperspectral object tracking," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2020, pp. 2106–2110.

[19] Z. Li, X. Ye, F. Xiong, J. Lu, J. Zhou, and Y. Qian, "Spectral-Spatial-Temporal attention network for hyperspectral tracking," in *Proc. 11th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Mar. 2021, pp. 1–5.

[20] M. A. Islam, J. Zhou, W. Zhang, and Y. Gao, "Background-aware band selection for object tracking in hyperspectral videos," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[21] M. A. Islam, W. Xing, and J. Zhou, "Multi modality Siamese feature fusion transformer tracker for object tracking from hyperspectral videos," in *Proc. 13th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Oct. 2023, pp. 1–5.

[22] W. Li, Z. Hou, J. Zhou, and R. Tao, "SiamBAG: Band attention grouping-based Siamese object tracking network for hyperspectral videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514712.

[23] C. Zhao, H. Liu, N. Su, C. Xu, Y. Yan, and S. Feng, "TMTNet: A transformer-based multimodality information transfer network for hyperspectral object tracking," *Remote Sens.*, vol. 15, no. 4, p. 1107, Feb. 2023.

[24] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[25] C.-Y. Wang, A. Bochkovskiy, and H.-Y.-M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7464–7475.

[26] L. Chen et al., "Object tracking in hyperspectral-oriented video with fast spatial–spectral features," *Remote Sens.*, vol. 13, no. 10, p. 1922, May 2021.

[27] Y. Tang, Y. Liu, and H. Huang, "Target-aware and spatial–spectral discriminant feature joint correlation filters for hyperspectral video object tracking," *Comput. Vis. Image Understand.*, vol. 223, Oct. 2022, Art. no. 103535.

[28] B. Uzkent, A. Rangnekar, and M. J. Hoffman, "Tracking in aerial hyperspectral videos using deep kernelized correlation filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 449–461, Jan. 2019.

[29] Y. Song et al., "VITAL: VIsual tracking via adversarial learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8990–8999.

[30] Z. Li, F. Xiong, J. Lu, J. Zhou, and Y. Qian, "Material-guided Siamese fusion network for hyperspectral object tracking," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 2809–2813.

[31] Y. Tang, H. Huang, Y. Liu, and Y. Li, "A Siamese network-based tracking framework for hyperspectral video," *Neural Comput. Appl.*, vol. 35, no. 3, pp. 2381–2397, Jan. 2023.

[32] Z. Liu, X. Wang, Y. Zhong, M. Shu, and C. Sun, "SiamHYPER: Learning a hyperspectral object tracker from an RGB-based tracker," *IEEE Trans. Image Process.*, vol. 31, pp. 7116–7129, 2022.

[33] J. Lei, P. Liu, W. Xie, L. Gao, Y. Li, and Q. Du, "Spatial–spectral cross-correlation embedded dual-transfer network for object tracking using hyperspectral videos," *Remote Sens.*, vol. 14, no. 15, p. 3512, Jul. 2022.

[34] Y. Tang, Y. Liu, L. Ji, and H. Huang, "Robust hyperspectral object tracking by exploiting background-aware spectral information with band selection network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[35] C. Sun, X. Wang, Z. Liu, Y. Wan, L. Zhang, and Y. Zhong, "SiamOHOT: A lightweight dual Siamese network for onboard hyperspectral object tracking via joint spatial–spectral knowledge distillation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521112.

[36] S. Wang, K. Qian, and P. Chen, "BS-SiamRPN: Hyperspectral video tracking based on band selection and the Siamese region proposal network," in *Proc. 12th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens.*, Sep. 2022, pp. 1–8.

[37] Y. Zhang, X. Li, F. Wang, B. Wei, and L. Li, "A fast hyperspectral object tracking method based on channel selection strategy," in *Proc. 12th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Sep. 2022, pp. 1–5.

[38] H. Nam and B. Han, "Learning multi-domain convolutional neural networks for visual tracking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 4293–4302.

[39] K. Cho et al., "Learning phrase representations using RNN encoder–decoder for statistical machine translation," 2014, *arXiv:1406.1078*.

[40] B. Li, W. Wu, Q. Wang, F. Zhang, J. Xing, and J. Yan, "SiamRPN++: Evolution of Siamese visual tracking with very deep networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4282–4291.

[41] X. Chen, B. Yan, J. Zhu, D. Wang, X. Yang, and H. Lu, "Transformer tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 8126–8135.

[42] Z. Chen, B. Zhong, G. Li, S. Zhang, and R. Ji, "Siamese box adaptive network for visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 6668–6677.

[43] D. Guo, J. Wang, Y. Cui, Z. Wang, and S. Chen, "SiamCAR: Siamese fully convolutional classification and regression for visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 6269–6277.

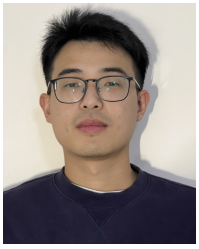
Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:57 UTC from IEEE Xplore. Restrictions apply.

[44] D. Guo, Y. Shao, Y. Cui, Z. Wang, L. Zhang, and C. Shen, "Graph attention tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 9543–9552.

[45] B. Yan, H. Peng, J. Fu, D. Wang, and H. Lu, "Learning spatio-temporal transformer for visual tracking," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10448–10457.

Mohammad Aminul Islam received the B.Sc. (Engg.) degree in computer science and engineering from the University of Chittagong, Chattogram, Bangladesh, in 2013, and the M.S. degree in computer science from Bangladesh Agricultural University, Mymensingh, Bangladesh, in 2019. He is currently pursuing the Ph.D. degree with the School of Information and Communication Technology, Griffith University, Nathan, QLD, Australia.

He is working as an Assistant Professor with the Department of Computer Science and Mathematics, Bangladesh Agricultural University. His research interests include spectral imaging, information fusion, computer vision, and pattern recognition.



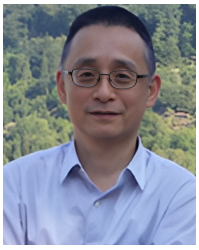
Wangzhi Xing received the M.S. degree in software development and the Diploma degree (Research) in information technology from Griffith University, Nathan, QLD, Australia, in 2021 and 2022, respectively, where he is currently pursuing the Ph.D. degree.

His research interests include image fusion, low-light object detection, and object tracking.

Jun Zhou (Senior Member, IEEE) received the B.S. degree in computer science and the B.E. degree in international business from Nanjing University of Science and Technology, Nanjing, China, in 1996 and 1998, respectively, the M.S. degree in computer science from Concordia University, Montreal, QC, Canada, in 2002, and the Ph.D. degree from the University of Alberta, Edmonton, AB, Canada, in 2006.

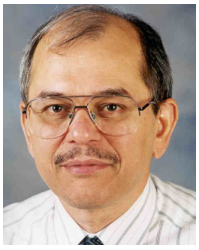
In June 2012, he joined the School of Information and Communication Technology, Griffith University, Nathan, QLD, Australia, where he is currently a Professor. Before this appointment, he was a Research Fellow with the Research School of Computer Science, Australian National University, Canberra, ACT, Australia, and a Researcher with the Canberra Research Laboratory, NICTA, Canberra. His research interests include pattern recognition, computer vision, and spectral imaging with their applications in remote sensing and environmental informatics.

Dr. Zhou is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and *Pattern Recognition Journal*.



Yongsheng Gao (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in electronic engineering from Zhejiang University, Hangzhou, China, in 1985 and 1988, respectively, and the Ph.D. degree in computer engineering from Nanyang Technological University, Singapore, in 2000.

He was the Leader of the Biosecurity Group, Queensland Research Laboratory, National ICT Australia (ARC Center of Excellence), Brisbane, QLD, Australia, a Consultant of Panasonic Singapore Laboratories, Singapore, and an Assistant Professor with the School of Computer Engineering, Nanyang Technological University. He is currently a Professor with the School of Engineering and Built Environment, Griffith University, Nathan, QLD, Australia, and where he is the Director of the ARC Research Hub for Driving Farming Productivity and Disease Prevention. His research interests include smart farming, machine vision for agriculture, biosecurity, face recognition, image retrieval, computer vision, pattern recognition, environmental informatics, and medical imaging.



Kuldip K. Paliwal was born in Aligarh, India, in 1952. He received the B.S. degree from Agra University, Agra, India, in 1969, the M.S. degree from Aligarh Muslim University, Aligarh, in 1971, and the Ph.D. degree from Bombay University, Mumbai, India, in 1978.

He has been carrying out research in the area of speech processing, since 1972. He was with a number of organizations, including the Tata Institute of Fundamental Research, Mumbai; the India Norwegian Institute of Technology, Trondheim, Norway; the University of Keele, Keele, U.K.; AT&T Bell Laboratories, Murray Hill, NJ, USA; AT&T Shannon Laboratories, Florham Park, NJ, USA; and Advanced Telecommunication Research Laboratories, Kyoto, Japan. Since July 1993, he has been a Professor with the School of Microelectronic Engineering, Griffith University, Nathan, QLD, Australia. He has co-edited two books *Speech Coding and Synthesis* (Elsevier) and *Speech and Speaker Recognition: Advanced Topics* (Kluwer). His research interests include speech recognition, speech coding, speaker recognition, speech enhancement, face recognition, image coding, pattern recognition, and artificial neural networks. He has published more than 300 articles in these research areas.

Dr. Paliwal is currently a fellow of the Acoustical Society of India. He has served as a Founding Member of the IEEE Signal Processing Society's Neural Networks Technical Committee, from 1991 to 1995, and the Speech Processing Technical Committee, from 1999 to 2003. He received the IEEE Signal Processing Society's Best (Senior) Paper Award for his paper on LPC quantization, in 1995. He was the General Co-Chair of the Tenth IEEE Workshop on Neural Networks for Signal Processing (NNSP2000). He was an Associate Editor of IEEE TRANSACTIONS ON SPEECH AND AUDIO PROCESSING, from 1994 to 1997 and from 2003 to 2004. He is on the Editorial Board of *IEEE Signal Processing Magazine*. He also served as an Associate Editor for IEEE SIGNAL PROCESSING LETTERS, from 1997 to 2000. He served as the Editor-in-Chief for the *Speech Communication Journal* (Elsevier), from 2005 to 2011.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:10:57 UTC from IEEE Xplore. Restrictions apply.